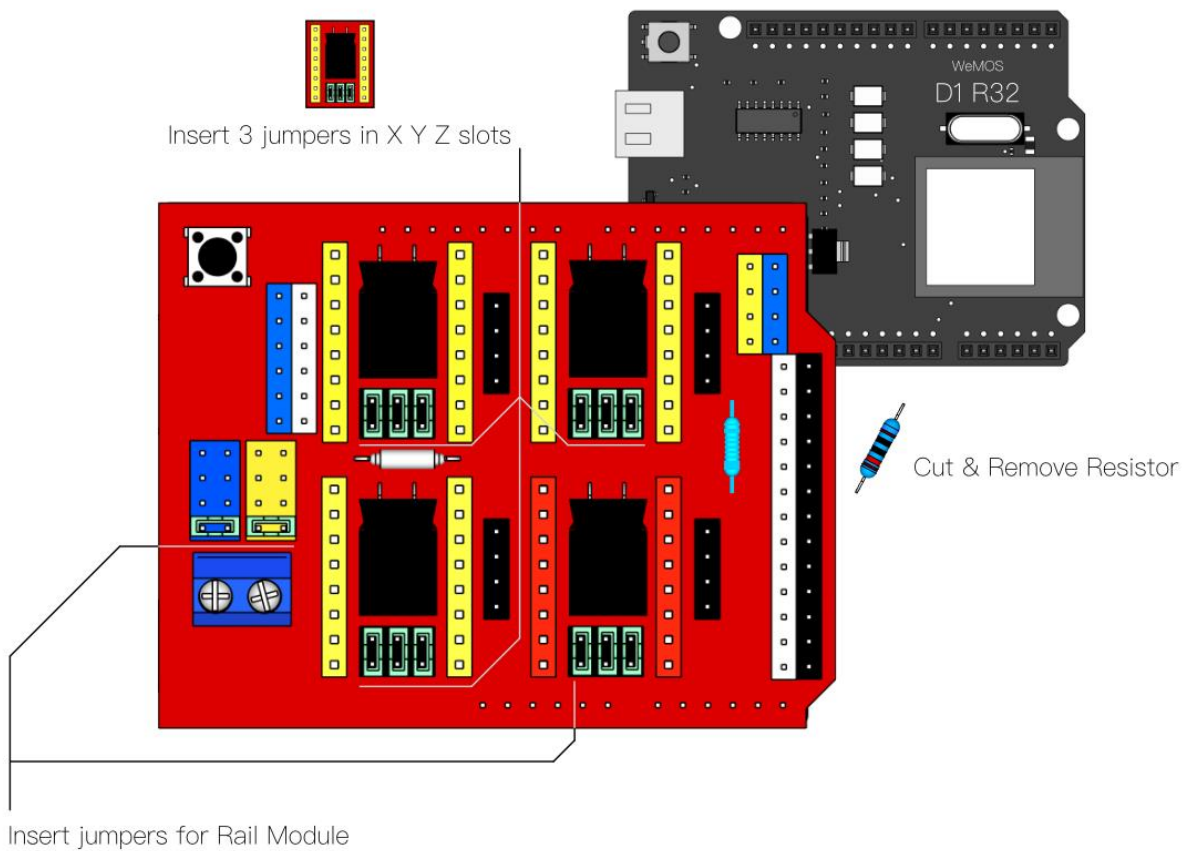
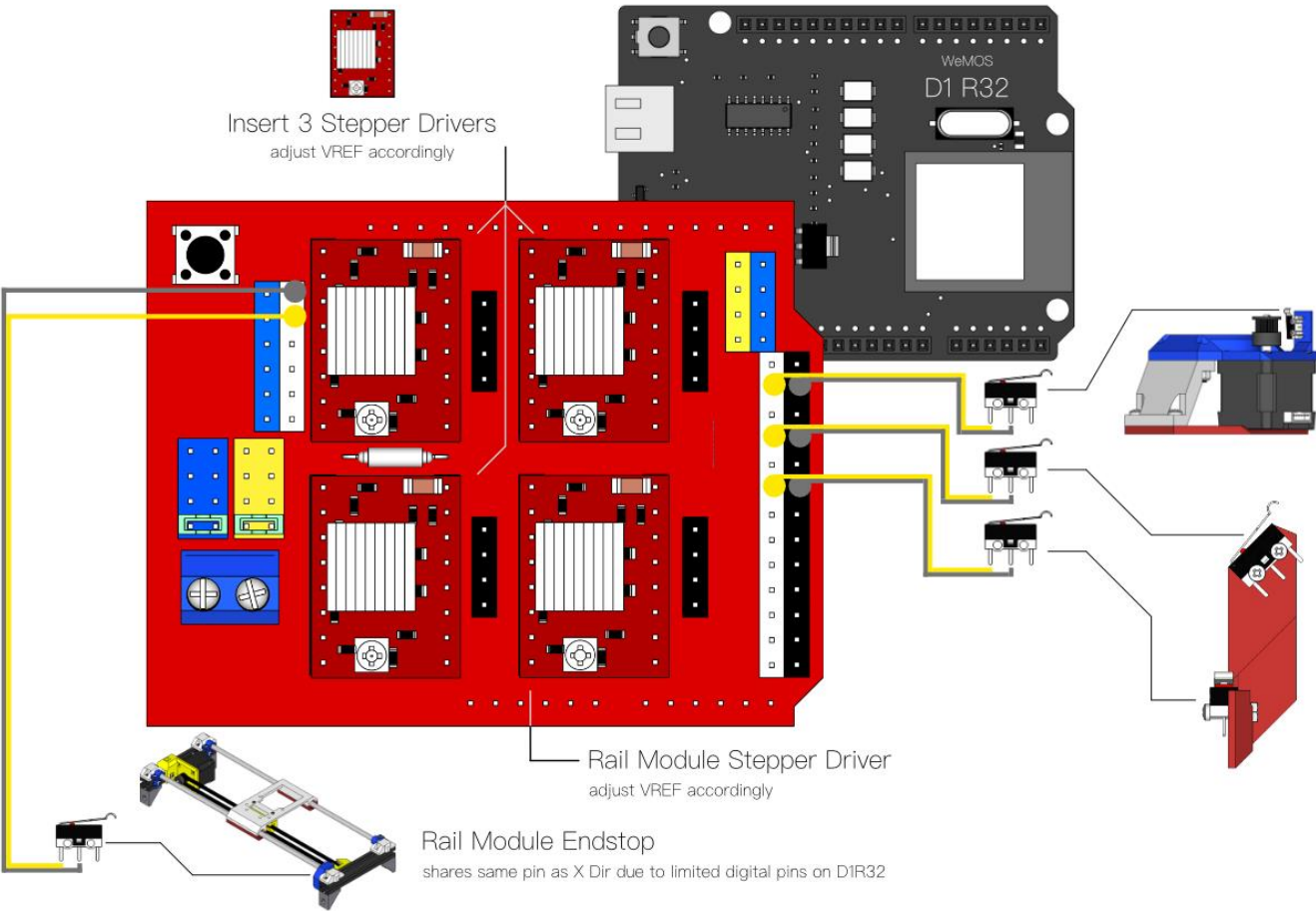


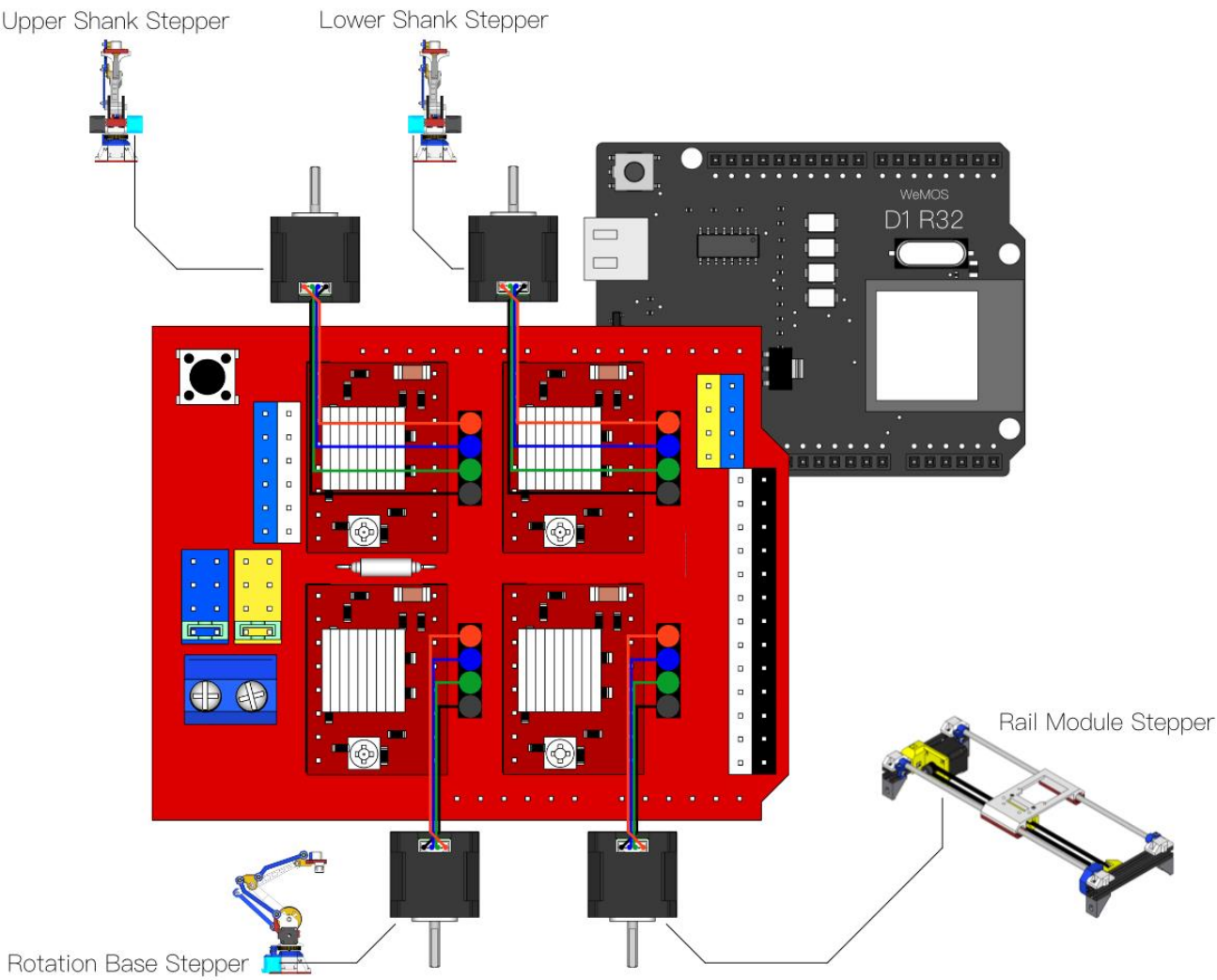
01 – ESP32



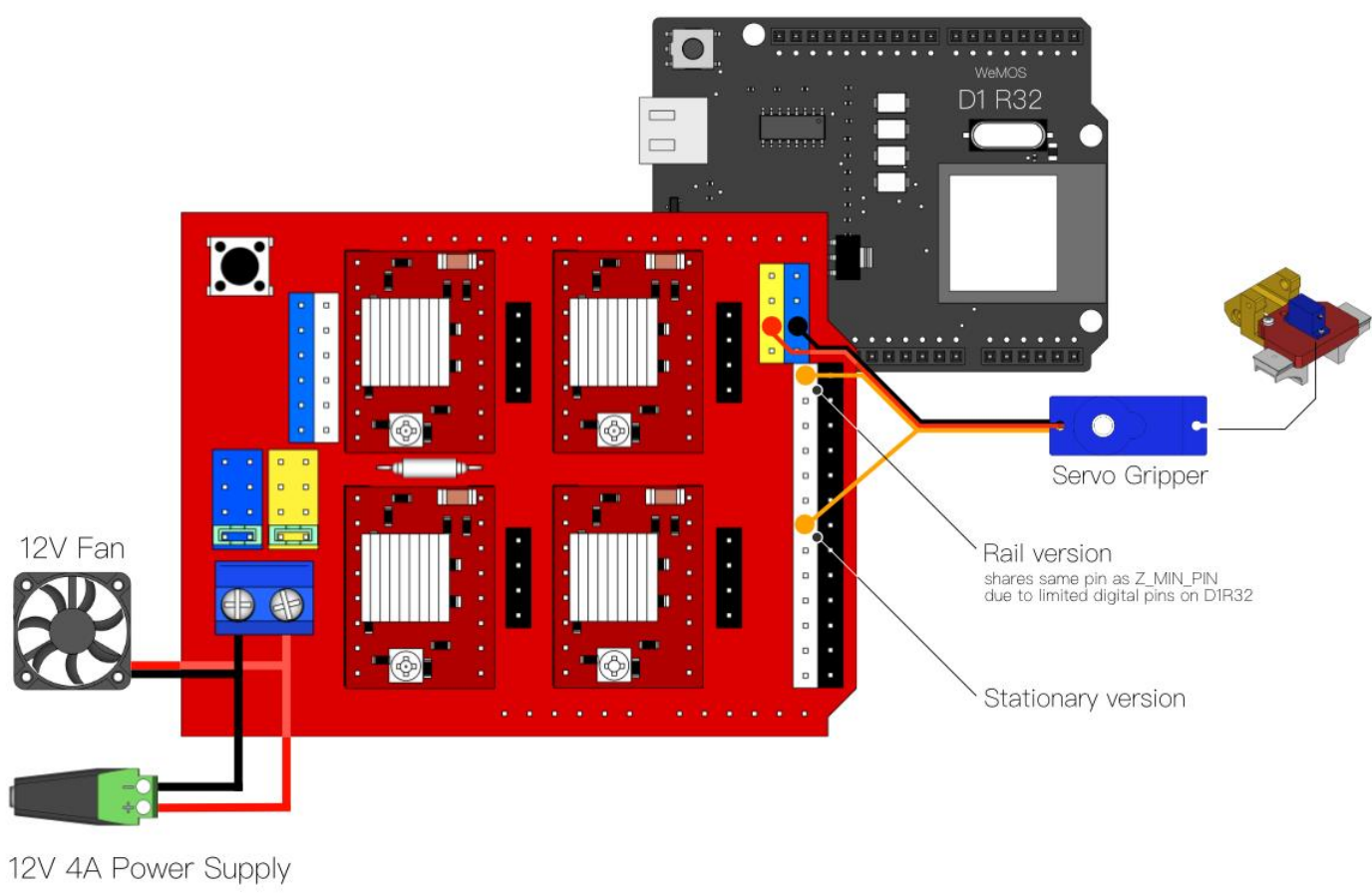
02 – ESP32



03 – ESP32

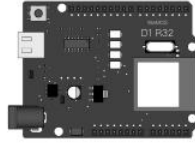


04 – ESP32



05 – ESP32

Arduino IDE Preparation (Part 1) WeMOS D1R32 (ESP32) Board



(MAC) Arduino > Preferences

(PC) File > Preferences

Additional Boards Manager URLs *

Add Link: https://dl.espressif.com/dl/package_esp32_index.json

Tools > Board > Boards Manager

Search 'esp32' & Select Version to Install

* Use Version 1.0.4 if 'PS4-esp32' Library is used *

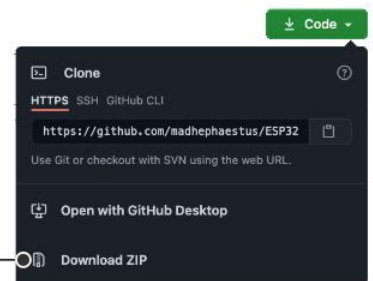
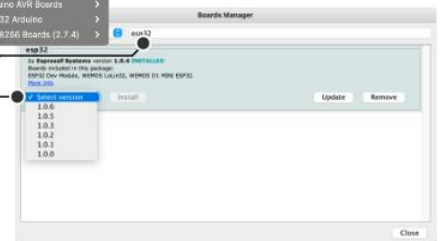
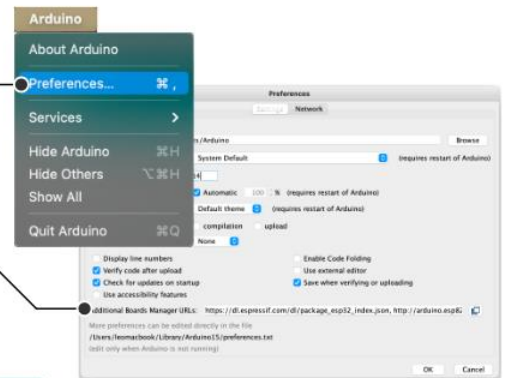
Download Dependent Libraries ZIP *

PS4-esp32 : github.com/aed3/PS4-esp32

ESP32Servo : github.com/madhephaestus/ESP32Servo

Sketch > Add .ZIP Library

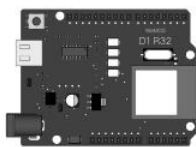
of downloaded library zip files



* Links & Versions subject to change. Refer to 20sffactory.com for latest revisions.

06 – ESP32

Arduino IDE Preparation (Part 2) WeMOS D1R32 (ESP32) Board



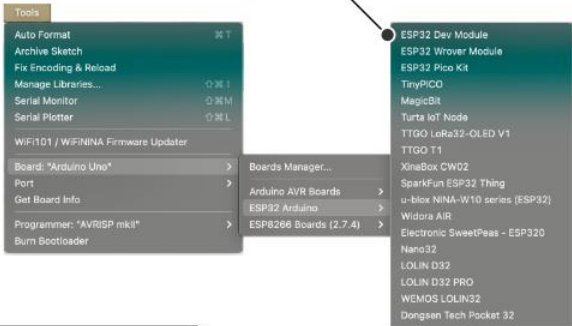
Set 'config.h' BOARD_CHOICE to WEMOSD1R32

supplement config guides:
20sffactory.com/robot/resource/firmware#config

```
#define BOARD_CHOICE WEMOSD1R32
```

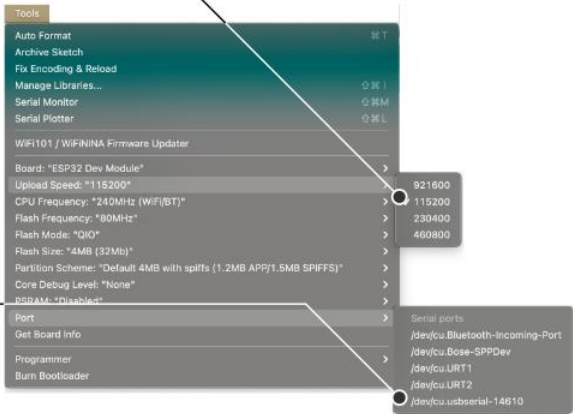
Select Board in Arduino IDE

Tools > Board > ESP32 Arduino > ESP32 Dev Module



Choose Upload Speed

115200



Connect Board via USB & Select Right Port

Compile & Upload
See "Command Guide" for Controls



07 – ESP32

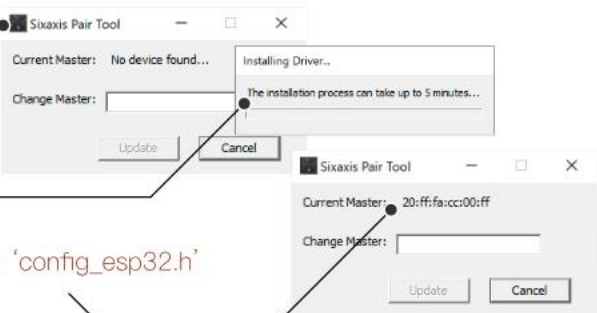
Configuration

Set `ESP32_PS4_CONTROLLER` in firmware `'config_esp32.h'` to true
Once configured true, Robot can only be online after Joystick is connected

```
#define ESP32_PS4_CONTROLLER true
```

Download Sixaxis Pair Tool on PC
sixaxispairtool.software.informer.com

Connect PS4 Controller to PC
Wait for Driver to Install



Ensure Matching Current Master & PS4_MAC in `'config_esp32.h'`
By Changing Master in Sixaxis Tool or Changing PS4_MAC value

```
#define PS4_MAC "20:ff:fa:cc:00:ff"
```

Control Buttons



08 – ESP32

Wiimote Joystick Robot Control Configuration V0.81

Set `ESP32_JOYSTICK` in firmware 'config_esp32.h' to `WIIMOTE`
Once configured, Robot can only be online after Joystick is connected

```
#define ESP32_JOYSTICK WIIMOTE
```

Change `JOYSTICK_SPEED_MULTIPLIER` to adjust control sensitivity

```
#define JOYSTICK_SPEED_MULTIPLIER 1.0
```

Upload Firmware & Power On

Hold Button 1 & 2 to connect Wiimote

After Connection Success

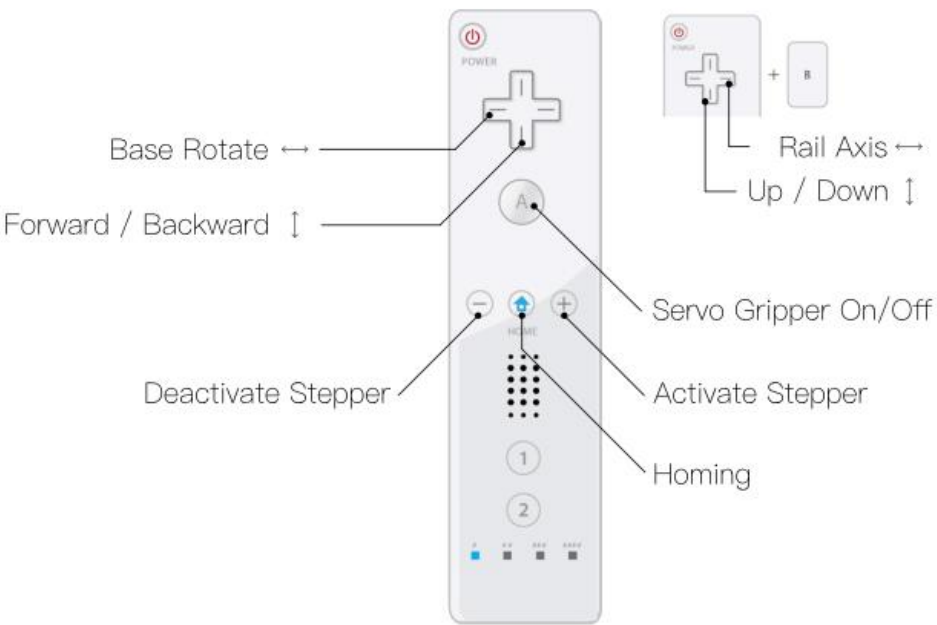
LED 1 Light turns on.

Serial Monitor displays:

INFO: WIIMOTE CONNECTED



Control Buttons



Base Rotate ↔

Forward / Backward ↑

Deactivate Stepper

Servo Gripper On/Off

Activate Stepper

Homing

Rail Axis ↔

Up / Down ↑